Claude + Mechanical Engineering Skills

Engineer

Voice

Text

“Design a light lattice for a 10 mm cell that is 2.5× stiffer vertically, printable in PLA.”

no tool training · no local models

Runs in Claude.ai, Claude Code and other Agent Skills hosts

Claude agent

understand · select · orchestrate · explain

loads a skill only when a request needs it

Design & CAD

Analysis & simulation

Manufacturing

32 skills · 19 package Barati Farimani Lab papers

SKILL.md
when & how to use

scripts/
tested tools compute

tests/ + notes
checks & provenance

Research papers packaged as skills

CADSmith · LinkD · RocketSmith · CFD Forge · StressD · GLU3D · LLM-SR · AdditiveLLM · + 11 more

Checked results

- **CAD models**
  STEP · STL
- **Simulations**
  FEA · CFD · flight
- **Process plans**
  G-code · process maps
- **Designs**
  lattices · linkages · rockets
- **Reports**
  calcs · plots · reviews

each with its evidence:
hand calc · exact case or re-simulation

Install once:
npx skills add
BaratiLab/mechanical-...